

Theory of Computation

Exercise 9: (Context-free grammar part 2)

1. Show that $L(G1) \neq L(G2)$.

$G1 = (\{S\}, \{a, b\}, \overset{V}{S}, \overset{T}{P1})$

$P1: S \rightarrow aSb \mid SS \mid \lambda$

SS
aSbSb
aaSbbab
aabbab

$G2 = (\{S\}, \{a, b\}, S, P2)$

$P2: S \rightarrow aSb \mid abS \mid \lambda$

S
aSb
aabSb
Can't derive

2. Find CFG for the language L.

$$L = \{ a^i b^j c^k : j = i + k \}$$

$a^i b^{i+k} c^k$

lambda
abbc, aabbbc, abbbcc

$S \rightarrow xy$

$x \rightarrow aXb \mid x$

$y \rightarrow bYc \mid \text{lamda}$

*3. Use CYK algorithm to find whether **abab** $\in L(G)$.

(Homework 7)

$$G: S \rightarrow AB$$

$$A \rightarrow BB$$

$$A \rightarrow a$$

$$B \rightarrow AB$$

$$B \rightarrow b$$

